Sheet Shrink: Efficient Spreadsheet Cleanup

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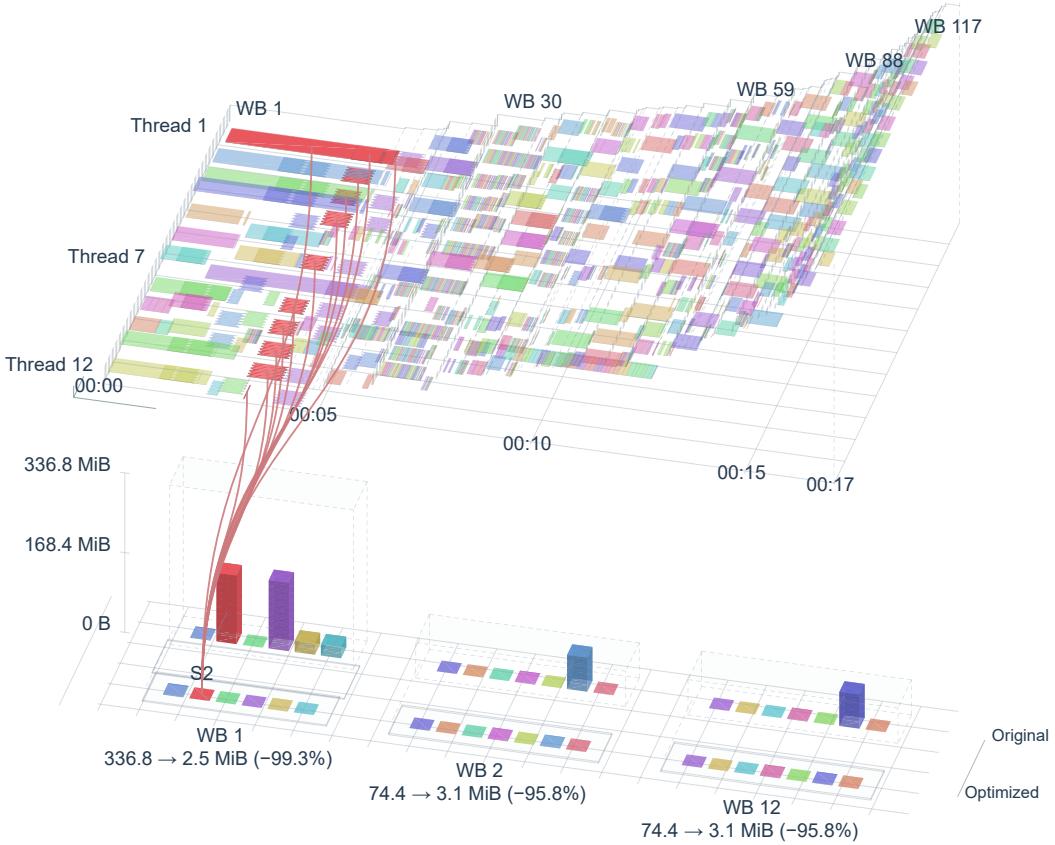


Fig. 1. Recorded development run on 144 Enron workbooks selected for large absolute XLSX reductions, using 12 workers. Work intervals follow workbook admission order (WB denotes workbook, numbered by admission). Thread labels identify worker lanes. The lower view compares the three largest uncompressed packages on one linear byte scale. Curves connect worksheet S2 in WB 1 to its work items. Observed wall time was 17.077 s.

***Abstract*—Spreadsheet packages can contain large populations of empty or redundant cell records that increase storage and ingestion costs. Cleanup must distinguish this structural redundancy from small formatting or dependency witnesses that preserve required workbook behavior. We present Sheet Shrink, a byte-oriented OOXML cleanup system that rewrites stored XLSX packages before opening them in Excel. It combines workbook-context analysis, local rewrite decisions, compact formatting representatives, and bounded parallel execution. We define its preservation contract and separate intentional policy choices from defects. Cleanup screening processed 1,182 Destatis, 8,623 Data.gov, and 30,301 prepared Enron XLSX inputs, reducing their aggregate compressed bytes by 15.48%, 5.29%, and 18.92%, respectively. Data.gov additionally had 494 unsupported-namespace skips. In a stress cohort of 147 workbooks chosen for large absolute reductions, 134 had targeted native comparisons against their selected originals and seven further observations used documented normalized baselines. Six lacked an eligible comparison. Independent native controls illustrate how retaining small formatting witnesses can preserve Excel's rendering. These measurements describe the selected capture scope.**

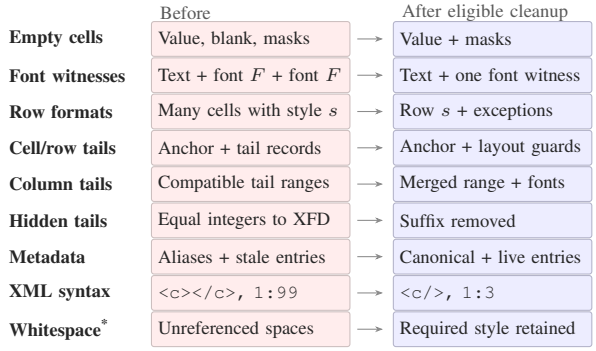


Fig. 2. Schematic cleanup families. Redundant cell formatting includes inherited fills and borders. Every pair has layout, protection and dependency guards. Row promotion requires compatible fixed-height context. Hidden-value removal admits guarded constant tails of at least 4,096 cells ending at XFD. *Whitespace removal is optional. Required witnesses and cell coordinates are retained.

I. Introduction

Spreadsheet files are both data containers and executable visual documents. A user can store a compact table in a workbook whose worksheet parts contain large numbers of otherwise empty cells. Such records can increase the work required to decompress and interpret the package. Removing every empty-looking record is unsafe. An empty cell can preserve a border, override an inherited fill, or influence automatic row height through its font. A whitespace value can prevent text from extending into its neighbour. Cleanup therefore requires a model of the effects that stored structures have on the document.

A motivating case is the Destatis migration workbook `wanderungen-2010120217005.xlsx`. In worksheet `Tabelle 3.4`, 21 rows each contain numeric zeros from column L through Excel's final column XFD. Those tails alone contain 343,833 stored values. They are hidden content, so removing them requires a stated admission policy and dependency checks in addition to empty-format cleanup. Section III-E describes the case and the values that the conservative policy deliberately retains.

For an XLSX input, Sheet Shrink makes these decisions from the stored package before loading the workbook in Excel. Its cleanup stage has no native rendering or calculated row-height observations. An Excel-hosted method can inspect the loaded layout and evaluate candidate edits against native observations. Our method instead retains compact source structures whose required effects follow from package analysis and the preservation rules. Native Excel is used later to evaluate the output. Legacy XLS conversion is a separately measured preparation stage.

Specialized spreadsheet parsing has demonstrated the importance of decompression, parsing, memory use and parallel execution for data loading [1]. Our problem adds a transformation obligation. The output remains an editable XLSX workbook and must retain required data, dependencies and visible presentation under a stated contract. File size alone is an inadequate measure of success. A smaller file that changes an important row height has failed that obligation.

Figure 2 summarizes guarded reductions across cells, rows, columns and workbook metadata. Sheet Shrink reuses existing style indices and compact witnesses to preserve required effects. Native saving can additionally compact style and shared-string tables.

The performance objective includes the complete user operation. For a sufficiently bloated workbook, we seek to make cleanup followed by opening the cleaned file faster than directly opening its original. This objective must be measured, including the cases where cleanup adds overhead. A single-thread comparison separates improvements in work performed from gains due to extra processing resources. Parallel corpus experiments then examine throughput and memory.

The paper develops the preservation model, the small-footprint mechanisms that implement it, and an evaluation combining structural checks with native visual observations. The evaluation distinguishes completed observations, unsupported inputs and unresolved differences. Historical Python implementations provide context for the engineering progression. Their different cleanup rules require separate output and correctness accounting in any timing comparison.

II. Preservation and Comparison Contract

A. Observable spreadsheet behavior

A spreadsheet package contains more than the populated table that a user sees. Individual cell records can carry formatting, protection, references, or values that affect other cells. Conversely, the presence of a stored record does not establish that it contributes to the intended presentation. Sheet Shrink targets production cleanup of such packages while retaining the structures needed by its preservation contract. Aggressive historical configurations are separate test artifacts and are excluded from the production subject of this paper.

The relevant visual area includes cells with meaningful values or formulas, empty cells with effective visible fills or borders, and graphical objects. Formatting inherited from rows or columns can contribute to this area. Conditional formatting and text extending into adjacent cells can also affect what is visible. The union of the original and transformed relevant areas is required when checking removed content. Layout changes in otherwise empty trailing rows or columns are permitted by the contract. Changes that move or alter relevant content remain subject to validation.

B. Information available before native loading

The cleanup decision uses stored values, formulas, styles, references and layout attributes. Native effective row heights and candidate before/after images are available only in the separate validation workflow. Stored custom heights can be retained directly. For automatically sized rows, a font-bearing blank can still require a representative even when its payload is empty.

An Excel-hosted alternative could measure a row, remove candidates and write the measured height as an explicit custom

height. SpreadsheetML provides a *customHeight* attribute for that policy [2]. Applying this technique requires a trusted height for the chosen renderer and a contract for replacing automatic sizing with a fixed height. The evaluated cleanup derives its compact preservation decisions from package context.

C. Removal decisions and compact preservation

A record can be eligible for removal only after accounting for its required contribution. An empty cell may still influence automatic row height through its font. A whitespace-only value may stop text from extending from an adjacent cell. Referenced hidden content can influence displayed values and charts. These cases require different reasoning from a test for an absent payload.

The preservation strategy retains a small representative when that representative can preserve a required effect. Row-font sentinels and guarded promotion of an existing style into a row format are examples. Their applicability depends on inherited formatting, gaps, merges, protection, and the available style records. An unresolved case retains the required source representation. The objective is to remove eligible bloat throughout the package while keeping the preservation footprint small.

The production policy admits separately guarded hidden constant tails and retains ordinary nonempty hidden tables, including unreferenced values. Tail admission depends on dependency and layout checks [3]. Absence of a reference alone does not authorize general hidden-data removal. A future Excel-hosted cleanup stage could measure native layout before replacing additional sentinels or whitespace blockers. Such a stage is outside the implementation evaluated here.

D. Validation and attribution boundaries

Our fast visual validation compares independently rebuilt original and Sheet Shrink states using the same installed Excel environment. Ordinary-save and rebuild controls retain information for attributing differences introduced by Excel. For XLS sources, conversion precedes the XLSX cleanup subject and has its own failure accounting. A file requiring repair or inaccessible protected content receives an explicit unsuccessful outcome.

Targeted visual equality is an observation over the selected regions. We evaluate the selection mechanism separately with native reference captures and mutations. Unequal pixels are retained for investigation. Repeated rendering controls, volatile dependencies, and capture geometry help classify their cause. The results section must report the completed coverage and unresolved findings at its fixed evaluation cutoff.

III. Method

A. Representation and analysis stages

Figure 3 summarizes the cleanup pipeline. The input is an XLSX ZIP package. The output retains its workbook structure and cell coordinates while removing records admitted by the preservation policy. Package metadata supplies style components, shared strings, relationships, merged regions and dependency sources. These contexts are prepared before cell-local decisions, so an index or an empty payload is never sufficient by itself to establish redundancy.

Require: Workbook package W, cleanup policy P
Ensure: Cleaned package or explicit unsuccessful outcome
1: Check supported serialization of W
2: Resolve styles, strings, layout and references
3: **for all** worksheets in W **do**
4: Classify content and effective formatting
5: Identify protected references and layout anchors
6: Select guarded removals under P
7: Retain compact font and geometry witnesses
8: Rewrite admitted records and retain other effects
9: **end for**
10: Reconcile names and calculation-chain changes
11: Assemble output with unaffected package members
12: Publish only a successfully completed package

Fig. 3. Sheet Shrink's logical cleanup stages. Guards are resolved before removal. Worksheet scheduling and chunking preserve these decisions.

Worksheet analysis consumes decompressed byte buffers without constructing an XML object for every cell. A state machine separates worksheet headers, column records, sheet data, merge information and the remaining suffix. In cell and row regions, bounded tag and attribute readers track hierarchy and quoting. Thus text resembling a coordinate inside a quoted metadata value cannot become a cell address. Small overlap state recognizes markers spanning input-buffer boundaries. Parsed metadata is retained only where later decisions require it. Names, references and other interpreted values can still require decoding. The implementation does not assume that arbitrary worksheet text is ASCII.

This organization reduces per-record allocation, but does not imply constant memory in worksheet size. The current implementation retains the sheet-data bytes, contexts and candidate metrics needed by subsequent passes. Decompression, rewriting, compression and retained analysis structures all contribute to peak memory. The supported input domain is normal unprefixed Transitional SpreadsheetML. Unsupported namespaces, encodings and ambiguous core structures follow explicit preservation or exclusion paths [3].

Figure 4 makes the worksheet preprocessor's control states explicit. The dispatcher separates column, cell and merge contexts. Read boundaries preserve partial state, and complete row prefixes can progress without consuming an unfinished row. Reconstruction follows successful arrival at the worksheet end.

B. Parser correctness and regression controls

Byte-oriented scanning transfers XML recognition and state-transition obligations to the implementation. Mature SAX and DOM libraries provide extensively exercised syntax handling, while application-specific scanners require explicit checks for element boundaries, quoted attributes, empty elements and incomplete input. Memory-safe execution still permits a scanner to misclassify content and emit an incorrect

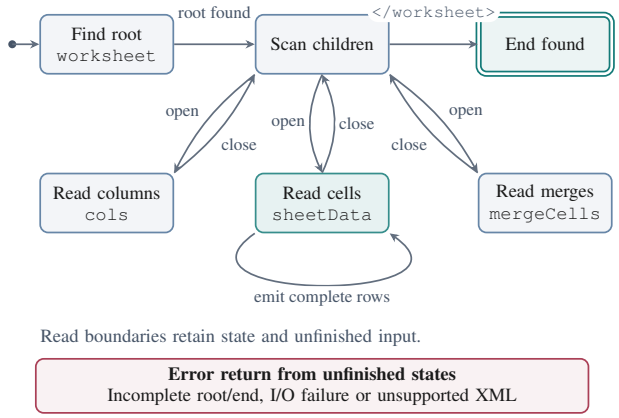


Fig. 4. The six worksheet-stream states. Child readers return to the same worksheet dispatcher. Row-aligned progress retains a partial row across reads. The lower box summarizes error outcomes. Reaching the end state permits reconstruction, with workbook publication subject to its own completion checks.

workbook. A rare serialization can expose such a defect after many successful inputs.

Our development history supplies a concrete example. A Data.gov workbook used `<v xml:space="preserve">` for visible string values. A scanner recognizing only the literal `<v>` classified these cells as empty and removed a heading and other strings. The correction recognizes valid element-name delimiters and retains attributed value and inline-string elements. Negative controls reject longer lookalike names such as `<value>` [4]. A separate defined-name regression ensures that a self-closing element cannot consume the following paired element during replacement [5].

Terminal-state checks are equally important. The worksheet stream must reach its closing state before output publication. An incomplete parse raises an error, removes partial output and preserves any existing destination. Integration fixtures exercise this failure after another worksheet has already been processed [6]. Streaming diagnostics additionally test comments, CDATA, quotes, nesting limits and source offsets across every input chunk size of their small fixtures. These diagnostic readers and the cleanup scanner have separate regression coverage.

At the time of writing, 31 correctness tests in the dedicated XML, namespace, shared-string, range, defined-name and streaming-reader modules exercise parsing and recognition behavior. A further 155 correctness tests in the cell, row and tail suites cover preservation witnesses, inheritance masks and content boundaries. Additional tests cover dependencies, styles, output construction and execution behavior. Counts refer to test functions, with parameterized fixture iterations counted once. The artifact records the source revision, test paths and counting criteria.

C. Effects and compact preservation witnesses

A numeric value, formula or inline string is content. In particular, zero and a shared-formula follower with no independent formula text remain content. An empty cell is classified by its effective formatting in the cell, row and column context. An explicit no-fill or borderless style can suppress a visible inherited effect. It remains necessary whenever deletion would expose that effect. Conversely, an explicit no-fill with nothing to clear can be redundant. Style components and application flags are therefore interpreted together.

Automatic row height introduces a different dependency. An empty cell's font can determine row height and move later content. Sheet Shrink retains a required font representative instead of every repeated occurrence. Equivalent font records can share an identity even when their style indices differ. Selection also respects local column-font masks and merged spans. A witness elsewhere in the row does not automatically replace a required local mask. Figure 5 isolates the mechanism in a constructed native Excel control. One witness preserves the observed height. Removing both moves the next visible row.

A fixed height eliminates this automatic-height effect, but an empty font-bearing cell can still affect the text baseline in a row containing text. The cleaner therefore distinguishes an empty fixed-height row from a fixed-height text row. Protection metadata does not erase a required font contribution. Protection formats inside the retained data extent also preserve interaction semantics, while separately guarded exterior tails can be removed.

Other witnesses preserve relationships between coordinates. Center-across-selection requires the contiguous blank span following the text cell. A borderless row can mask column borders, and an empty height-bearing row can determine their vertical extent. Column widths remain necessary when they support visible row borders. A chart can inherit a number format from an empty first series cell. These structures extend the preservation problem beyond the last nonempty value. Figure 6 shows an Enron counterexample in which Check Performance's smaller output exposes a different inherited fill. The observation concerns the pictured rebuilt region.

D. Guarded promotion and tail pruning

A second compaction replaces repeated cell styles with an existing row format. The row must be visible, unprotected and without an existing row style. Both a stored height and an enabled `customHeight` flag are required. A stored `ht` alone does not freeze automatic sizing. Content cells, merged cells, missing positions and inherited column formats must remain compatible with the proposed row format. Figure 7 illustrates the full-width case [7].

The dominant empty-cell style occurs at least twice. Selection counts unsuitable styles too, then applies eligibility checks. Otherwise an eligible minority could incorrectly replace an unsuitable majority. Ties choose the smaller existing style index. Dense promotion requires at least 1,024 occurrences. A bounded-gap route admits at most 64 missing columns and restores required original formats only under its inheritance checks. Sparse promotion requires equivalent inheritance at every absent position and excludes merged sparse rows. Uniform-fill promotion additionally checks content-cell

One font witness preserves row layout

A2 has Arial 10. Empty B2 and C2 have Arial 18.

Source

```
 <row r="2">
  <c r="A2">...</c>
  <c r="B2" s="1"/>
-  <c r="C2" s="1"/>
 </row>
```

Guarded result

```
 <row r="2">
  <c r="A2">...</c>
  <c r="B2" s="1"/>
 </row>
```

Original control **23.25 pt**

Font witness control			
10 pt text			
Next visible row			

One witness retained **23.25 pt**

Font witness control			
10 pt text			
Next visible row			

Both witnesses removed **12.75 pt**

Font witness control			
10 pt text			
Next visible row			

Native Excel vector export after full rebuild. Removing the repeated witness is safe here. Removing both moves the next visible row.

Fig. 5. A reduced font-witness control. Red marks the removed duplicate. The source and one-witness native vector exports match after a full rebuild in Excel 16.0 build 20326. Removing both empty Arial 18 cells reduces row 2 from 23.25 to 12.75 points and moves the following row. XML is abbreviated. These are constructed native print controls with gridlines, separate from the corpus regression.

fills. Unknown metadata or style identities prevent promotion and retain the ordinary cleanup path.

Tail pruning removes formatting-only structure beyond retained content and layout anchors. Font-only blanks after the last relevant anchor need not preserve the position of nonexistent later content. Visible fills, borders, conditional regions and drawing anchors can still extend that boundary. A white row fill is redundant only when compatible column fills cover every column through XFD. An uncovered column breaks that argument. Compatible extraneous column ranges can be coalesced, but local overrides and border-supporting geometry remain.

E. Hidden content and the Wanderungen case

Ordinary hidden nonempty tables remain intact. The separate constant-tail policy admits only a dense horizontal suffix of at least 4,096 identical signed integers, hidden and ending at XFD, subject to cell-level layout and dependency guards. Gaps, formulas, text, differing values and unsupported numeric forms reject the band. Source cells contribute before style cleanup. The common zero path compares value bytes directly, while one admitted row span replaces per-cell entries in the later reference map. Temporary row-local candidates still exist.

The motivating migration workbook contains 21 such zero tails in `Tabelle 3.4`: rows 11–30 and 32, from L through XFD. Each contains 16,373 zeros, totaling 343,833. Source ZIP inspection confirms these counts and retains the source hash. The historical cleanup investigation records 376,573 removed zeros across two worksheets, alongside 243 preserved nonzero values and a retained short 237-zero tail [8]. The full abbreviated XML example is included in the companion atlas. These are content removals under an explicit structural policy. A deliberate constant full-width matrix can meet the same heuristic, so repetition alone does not establish user intent.

F. Workbook dependencies and deferred rewriting

Reference-sensitive candidates remain pending until workbook-level analysis has seen supported formulas, names, chart and object formulas, and pivot dependencies. A formula on a later worksheet can otherwise refer to content already removed. References resolve through workbook relationships and sheet scope. Shared-formula followers move relative axes while retaining absolute axes. The implementation protects enclosing source rectangles without expanding each follower cell. Dynamic addressing, unresolved scope and unsupported expressions prevent the required deletion argument [9].

A known reference intersecting an admitted hidden span protects the entire span. Only admitted candidates reach rewriting. Removing a payload need not remove its format if that format still masks an inherited effect. After deferred changes, affected rows lose their stale tail anchors and pass through cleanup again. Otherwise records retained solely because of the removed anchor would survive incorrectly. Whitespace removal is disabled in Default because whitespace can control text overflow and row layout.

Package metadata has separate predicates. Calculation-chain cleanup resolves removed cells by sheet identity and preserves unresolved mappings. The cleaner does not evaluate formulas or manufacture replacement caches. Built-in alias cleanup currently matches name and local scope without proving equality of the two formula bodies. This is a specific limitation of that policy.

G. Parallel work and output ownership

Figure 8 relates the task hierarchy to a schematic execution across threads. Worksheet identity remains shared across its

An empty gray cell masks an inherited pink fill

Column B uses XF 51 (pink). Cell B31 uses XF 98 (gray).

Original

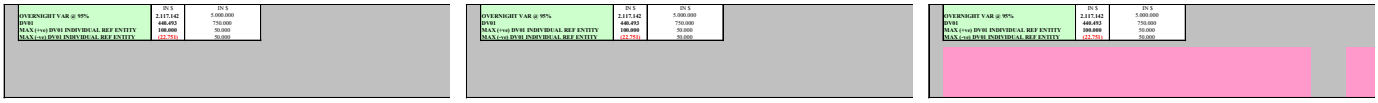
```
<c r="B31" s="98"/>
```

SheetShrink Default

```
<c r="B31" s="98"/>
```

Excel Check Performance

```
<!-- B31 removed -->
```



PL-Consolidated!A25:H37, vector export after full rebuild. Original and Default match pixel-for-pixel here. Removing the empty gray cells exposes the inherited pink background.

Fig. 6. Empty cells can preserve a visible fill. In Enron case `ccde4294bab9094d7d73`, an explicit gray B31 cell masks a pink column format. The retained Original and Sheet Shrink Default vector exports match in `PL-Consolidated!A25:H37` after full rebuild of the retained rebuild/Save controls. Check Performance removes the mask and exposes pink. The source excerpt is shortened. The print-renderer equality observation covers this region.

chunks, while ordered assembly separates execution order from output order. Workbook, worksheet and row-chunk work share the cleanup execution budget. A permit limits active workbooks and remains held through output finalization. Workers can process other workbooks while one long worksheet remains active. Large worksheets expose row-aligned chunks to Rayon, allowing idle workers to help without splitting a row's font, promotion or tail decisions. The chunk planner estimates byte-sized work, then adjusts boundaries to complete rows. Small ranges use the serial path. Each chunk receives isolated metrics, and results are assembled in source order.

Ready chunks participate in admission decisions for new worksheets. A smaller sheet can yield admission while larger chunk work is pending. This priority is an implementation policy over the common work-stealing pool, not a distinct semantic rule. Batches retain only a bounded number of chunk results while the assembled output grows. Workbook contexts, the output buffer and deferred candidates remain additional memory consumers. Thread count, active-workbook count and writer count are separate experimental parameters.

Changed worksheet members are precompressed in parallel and handed to a workbook writer. The writer owns ZIP mutation and publishes a completed partial output only after processing succeeds. Unaffected members can reuse compressed bytes. If all entries are original, the source workbook is copied byte for byte. A fast unchanged-entry classification can use CRC32 and uncompressed length. That shortcut is collision-prone, and exact byte verification is available. Preservation and timing experiments must record which identity policy was used. Cryptographic evidence reuse is a separate validation mechanism.

The method retains existing style and shared-string indices rather than globally renumbering their tables. This avoids unrelated rewrites and permits copy-through of unaffected members. It also leaves reductions requiring native save or global table compaction outside the cleanup subject. Work events retain the relation between workbook, worksheet and chunk jobs for diagnosis and replay. The UI is an observation tool, not evidence that unobserved regions are unchanged.

H. Native validation and artifact scope

DeltaSieve compares selected potentially affected regions after native rebuilding of separate original and transformed copies [10]. Save-only controls distinguish Excel serialization from the cleanup operation. Independent reference plans and visible mutations evaluate the selector separately. Agreement of selected images alone cannot establish whole-workbook preservation.

The companion atlas provides 50 abbreviated source/result examples, negative cases and pinned implementation anchors at figures/xml-methods/xml-cleanup-atlas.pdf. It includes the Wanderungen XML and the fixed-height, protection and dependency cases summarized above. The examples omit irrelevant wrappers and repeated cells, with local style indices. Native controls are distinguished from schematic illustrations. Disabled optional paths and the inactive row-attribute helper are identified in its coverage index.

IV. Evaluation

The evaluation asks three questions. RQ1 examines compressed package reduction and the remaining preservation footprint. RQ2 measures cleanup time, memory and the combined cleanup-plus-opening operation. RQ3 examines structural preservation and observed native presentation under the stated validation scope.

A. Corpus identity and outcome accounting

The evaluation treats Data.gov, Destatis and Enron as distinct populations. Archive checksums, member names and duplicate occurrences identify each selected corpus version.

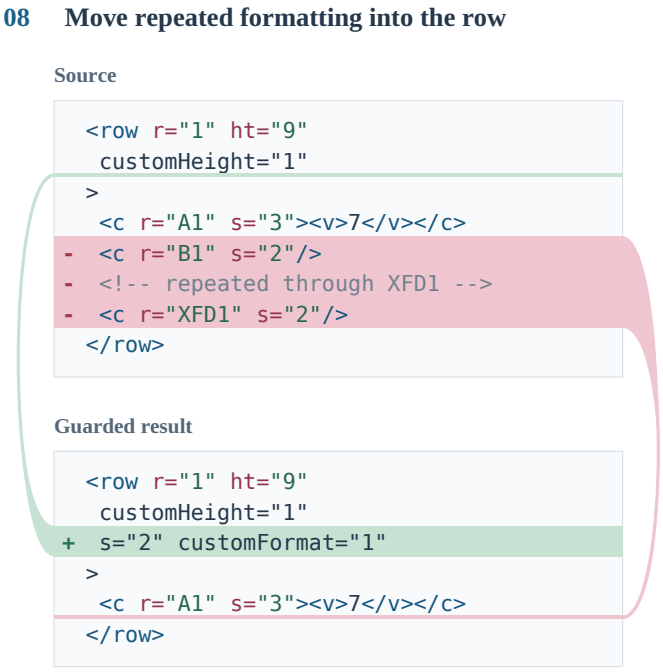


Fig. 7. Guarded dense-row promotion reuses an existing XF and keeps the value cell. Red marks removed lines and green marks added lines. The illustration omits the repeated middle cells. The full-width regression removes 16,383 empty cell records under the fixed-height and inheritance guards.

Legacy XLS conversion is reported separately from XLSX cleanup. A historical Enron development selection contains 20,872 inputs, with 20,503 prepared artifacts and 369 unsuccessful preparation outcomes at its recorded preparation checkpoint. Its selected visual scope contains 16,799 workbooks. These counts describe that development selection and do not establish completed visual coverage. The replacement EDRM Enron v2 XML handover contains 34,065 distinct content-addressed candidates totaling 8.470 decimal GB of uncompressed payload. The imported selection unit is a distinct candidate SHA-256. Candidate transport was checked against the supplied manifests, and the import retained the original candidate bytes. This externally generated extraction includes legacy Excel and other OLE candidates. Native format recognition, successful XLSX conversion and visual eligibility remain separate evaluation stages. Candidate count therefore describes the imported input frame, with workbook outcomes to be reported after processing. The upstream extraction provenance and local import identities are retained alongside the experiment.

The current native visual stress cohort selects the 49 whole workbooks with the largest absolute compressed-file byte

TABLE I
FROZEN XLSX SOURCE INVENTORIES. SIZES ARE DECIMAL GB OF LISTED WORKBOOK FILES AND DO NOT MEASURE COMPLETED NATIVE COVERAGE.

Corpus	XLSX members	XLSX total (GB)
Data.gov	9,117	25.273
Destatis	1,182	2.364

reduction in each corpus, totaling 147 original identities. All processor baselines use those same identities. Earlier development observations outside this reduced cohort remain retained history. Availability of an optimized variant and completion of its visual comparison are reported separately.

The two frozen XLSX inventories in Table I come from the versioned source profiles. Each listed archive member has an individual SHA-256 identity. Their source-frame sizes are separate from the eligible, completed and independently referenced subsets.

A dated evaluation cutoff will report compared workbooks, differences, failures, unsupported representations and unfinished work. Repeated attempts have their own counts. An unchanged package can bypass visual work under an exact input-identity check, but a failed capture cannot contribute an equality observation. The full recursive Enron XML-derived corpus is a separately identified population.

B. Cleanup screening outcomes

Fresh release-build runs produced 1,182 Destatis, 8,623 Data.gov, and 30,301 prepared Enron XLSX outputs. Across those outputs, aggregate compressed file bytes fell by 15.48%, 5.27%, and 18.87%, respectively. The smaller, equal-size, and larger output counts were 1,045/30/107 for Destatis, 4,755/3,522/346 for Data.gov, and 21,565/8,415/321 for Enron. Hash checks confirmed byte identity for all equal-size outputs in these runs. Data.gov also had 494 unsupported-namespace skips. With sixteen workbook slots and sixteen Rayon threads, elapsed times were 119.7, 1,134.5, and 181.6 s, respectively, including analysis statistics and full replay events alongside concurrent native validation [11]. All three runs reached completion with zero reported processing failures. These namespace exclusions occurred only in Data.gov in the evaluated runs. These descriptive results include files that grew and use only processed input bytes as the denominator. Enron's denominator begins after preparation. Visual preservation is evaluated separately. Figure 9 shows the workbook and worksheet distributions for these runs. Empty-cell cleanup and trailing layout rows account for prominent worksheet XML reductions. The contribution of column range merging is especially visible in Destatis. The worksheet panels exclude 88 prepared Enron workbooks without worksheet XML observations. Workbook payload sizes use all uncompressed ZIP member sizes. A separate inventory of the 9,117 Data.gov inputs identified 493 workbooks with explicitly prefixed worksheet data elements. Experimental copies underwent namespace normalization followed by Default cleanup, with savings measured against the normalized baseline. Only

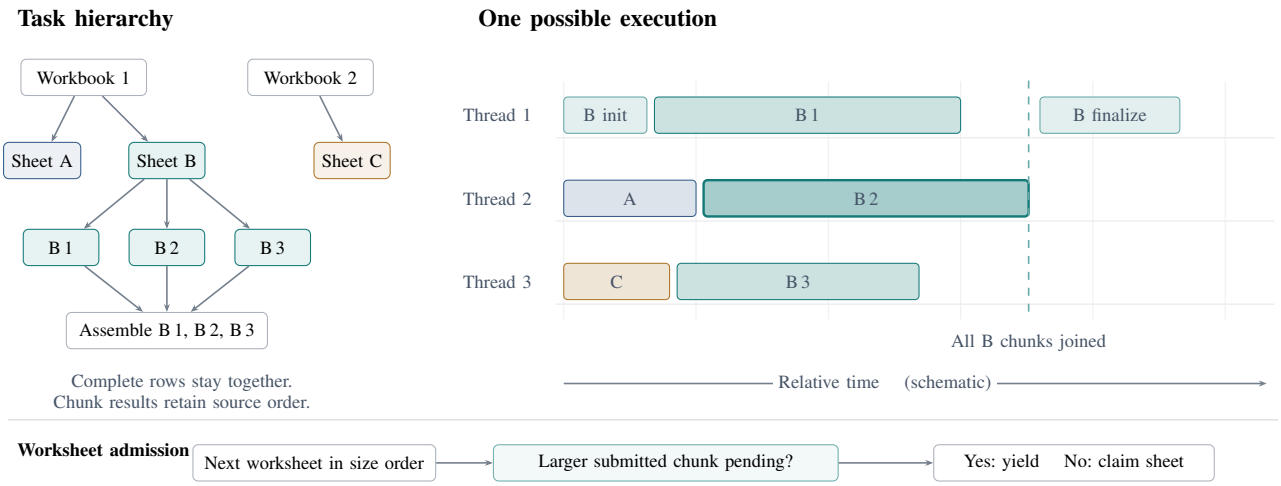


Fig. 8. Hierarchical parallelism in Sheet Shrink. Workbook admission bounds active inputs, while worksheet tasks and row-aligned chunks use the common Rayon pool. Matching colors connect the hierarchy to an illustrative thread schedule. B 1–B 3 belong to one worksheet and can execute on different threads. Their results are assembled in source order. A submitted larger chunk can defer a smaller worksheet. Bar lengths are schematic. Individual steal events are not recorded by the current work observer.

two produced smaller compressed packages, with reductions of 6.415% and 0.047%. Other outputs could lose redundant cell records without becoming smaller as ZIP packages. The observed compressed-size benefit was therefore concentrated in two inputs [12].

Earlier exploratory inspection associated prefixed serializations with generated exports. The subsequent inventory found a maximum row coordinate of 694,981 across the 493 matches, within Excel's row limit. Producer attribution and application compatibility require additional evidence. Excel itself supports Strict Open XML output [13]. Production eligibility checks use bounded package-root headers and declarations encountered during streaming, allowing unsupported representations to be identified before native opening. The measured opportunity supports retaining the current parser scope and reporting excluded inputs explicitly, with the experimental normalized copies available for further investigation.

Corpus-driven development exposed both recognition and preservation defects. The Data.gov attributed-value failure is described in the method, and the Destatis automatic-height correction is examined below. Earlier Enron controls isolated deleted center-across-selection spans and fill masks by restoring only the affected original cell records. These findings motivate retained fixtures and native controls alongside the screening counts [14].

C. Sampling and reference controls

The random evaluation sample is selected from a frozen eligible frame with a recorded seed before inspecting visual findings. Strata use corpus identity, package size and structural features available before validation. Low-reduction and unchanged files remain represented. A separately labelled development stress set contains large and historically problematic workbooks.

Feasible sampled files receive full native reference captures over the relevant visual area using a capture plan independent of the candidate selector. A full-reference timeout remains in the denominator. It is not replaced silently by an easier workbook. Mutations with known visible effects cover cell values, styles, geometry, conditional dependencies and objects. Mutations that Excel normalizes away are reported separately.

D. Performance and comparison subjects

We compared direct open–save trials with fresh six-worker Sheet Shrink cleanup followed by the same library. Across three stress workbooks, all 48 Sheet Shrink trials completed under a 4 GiB commit limit, compared with 33 original trials. Median elapsed time improved in ten of eleven fully completed pairs. Migration with openpyxl took 111.3 s directly and 9.7 s including cleanup. The accompanying benchmark receipts retain the individual phases and the second-pass measurements.

Tables II and III join the opening/memory case study through frozen file identities. Migration, Municipal Finance and QT/RR have 288 retained opening/saving attempts across six variants and three repetitions. The selected Enron workbook has a separate eighteen-attempt follow-up on the larger-memory host, described below. Its measurements retain their own host and configuration. The size table checks the exact retained comparator outputs against their manifest hashes. Fresh Sheet Shrink outputs have recorded compressed sizes, while their retained uncompressed-member inventory and matched native evidence remain pending. Two exact Municipal Finance comparator pairs have completed native reports with scope limitations. Their diagnostic pixel observations are retained alongside affected worksheets and geometry findings. Total comparable-pixel denominators remain pending. The artifact contains source identities, report hashes, per-region diagnostic counts and all trial receipts.

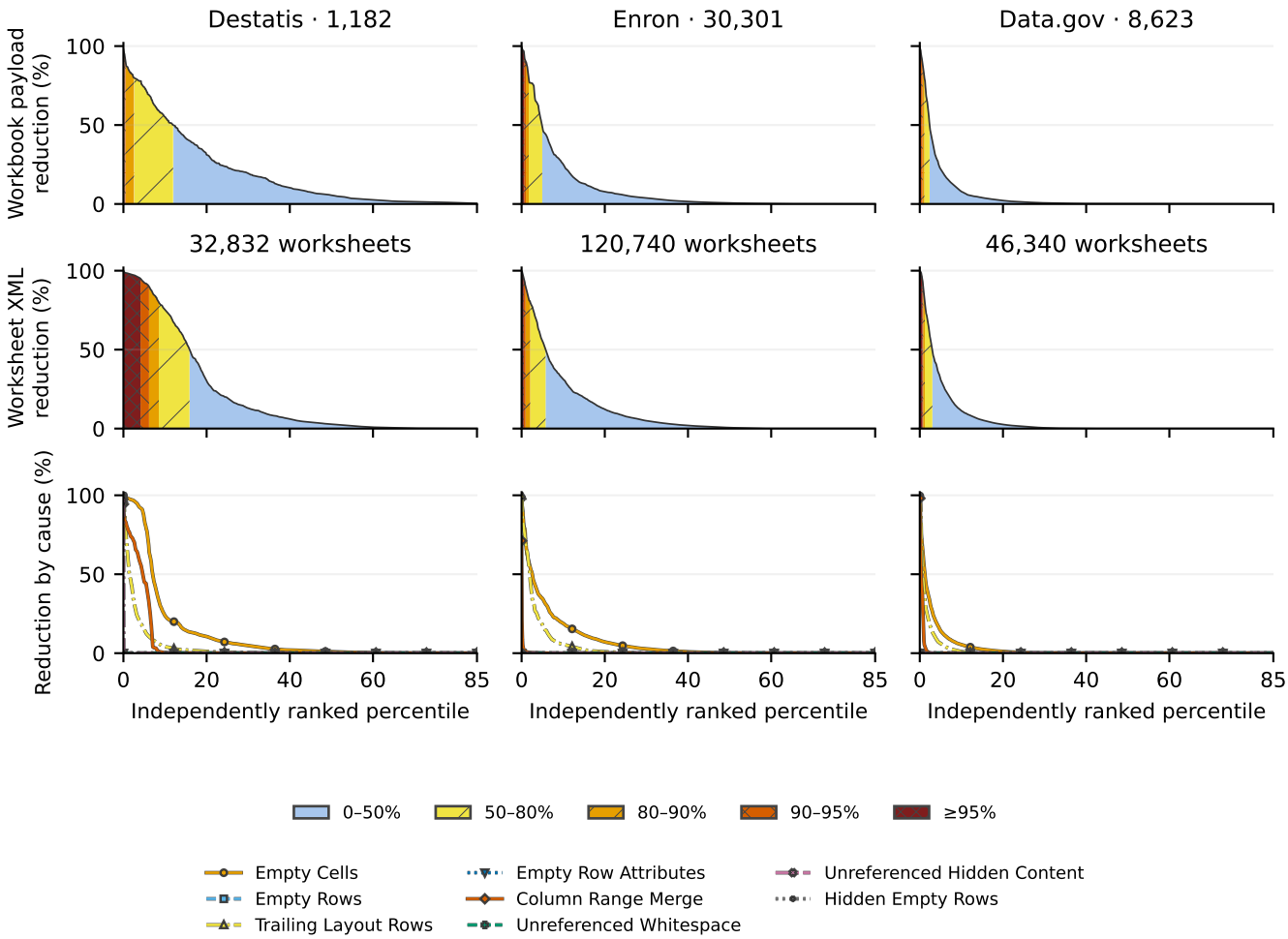


Fig. 9. Full-corpus uncompressed-size reduction. Top, complete workbook payload from all ZIP member sizes recorded in the central directory. Middle, uncompressed worksheet XML. Bottom, worksheet XML reduction attributed to each recorded cleanup cause. Each curve independently sorts its observations from greatest to least reduction. The x-axis gives the fraction of observations traversed in that ordering. The common display ends at the 85th percentile. Remaining positive reductions are below 0.5 percentage points. The omitted tail includes unchanged outputs and growing workbooks (3/24/5) and worksheets (82/143/39), in Destatis/Enron/Data.gov order, retained in the accompanying data. Colors and hatching mark 50/80/90/95% reduction tiers. Data.gov's 494 unsupported-namespace inputs are excluded from these processed-output distributions. The worksheet panels cover 1,182 Destatis, 30,213 Enron and 8,623 Data.gov workbooks with recorded worksheet XML. Enron begins with prepared XLSX inputs.

The resource table separates physical working-set samples from the job's committed-memory peak. Each displayed time is a median of completed attempts, with memory maxima over all three attempts. Stopped attempts retain their elapsed time and memory boundary. CPU totals, individual timings and ranges remain available in the machine-readable artifact. Existing prepared-output variants measure subsequent library processing. Fresh Sheet Shrink rows include cleanup within the measured pipeline. The common host is a Ryzen 5 4500U laptop with six logical processors, 8 GiB installed RAM and Windows 11. Each job has a 4 GiB commit cap and a 900 s timeout, with the free-RAM reserve disabled. The Sheet Shrink executable is frozen at revision `765c480` and uses six Rayon and six writer workers. Native visual observations use the retained policy-v12 paired-rebuild protocol. Later software revisions and other hosts require their own identified measurement lanes.

E. Larger-memory-host follow-up

The five workbook/library combinations that reached the laptop's 4 GiB commit ceiling were repeated on an AMD Ryzen 7 7800X3D host with sixteen logical processors and 102,066,212,864 OS-reported physical bytes. Each combination has three Original and three fresh Sheet Shrink pipeline attempts at both 4 GiB and 24 GiB aggregate process-tree commit limits, totaling sixty attempts. The same frozen Sheet Shrink binary uses sixteen Rayon and sixteen writer workers with one active workbook. The controller maintains an 8 GiB reserve for physical RAM and commit headroom, with a 900-second per-trial timeout. Python 3.12.14, openpyxl 3.1.5, pandas 3.0.1, R 4.6.2, readxl 1.6.0 and writexl 2.0.1 identify the tested software lane [15].

At 4 GiB, all fifteen Original attempts reached the memory limit, while all fifteen fresh-cleanup pipelines completed. At 24 GiB, the nine Original attempts for Migration/readxl, Finance/openpyxl and QT/RR/openpyxl completed. Migration's two pandas endpoints reached the memory limit in all six attempts. All fifteen fresh-cleanup pipelines completed at that limit. Table IV retains these censored endpoints alongside successful medians. A memory-stop termination time receives no completed-runtime ratio.

The larger limit distinguishes memory censoring from completed processing cost for these selected cases. Native visual comparisons, a separate cleanup benchmark and read-only evidence analysis ran concurrently. Filesystem caches were retained. The measured ratios are descriptive paired results under that context. This host lane retains its own configuration and observations alongside the earlier six-worker laptop study. Across all sixty trials, sampled physical RAM and commit headroom remained above the declared system reserve.

After the timing series, separately regenerated controls for all three inputs matched every uncompressed member name, length and SHA-256 of the retained native candidate identities. These representative controls contain 138, 225 and 24 members for Migration, Finance and QT/RR. Migration's native observation uses a documented normalized baseline, and QT/RR's retained selected plan has zero captures. The full per-trial member inventories were not retained for the timed fresh outputs, whose archive hashes remain in the receipts. These qualifications delimit the native/resource association. A second cleanup was measured after each completed fresh pipeline, outside the timed endpoint. Its recorded ZIP identity flags do not establish general cleanup idempotence.

F. Enron selected-case follow-up

The fourth case uses the Enron workbook whose original SHA-256 begins `7298df9a6a76`. Its original archive contains 42,710,945 compressed bytes and 353,151,842 uncompressed member bytes. On the same larger-memory host, six variants each received three openpyxl 3.1.5 round-trip attempts, with a 4 GiB process-tree commit limit, an 8 GiB physical-RAM and commit reserve and a 300-second timeout [16]. All eighteen attempts have terminal receipts. Original and XLStylesTool reached the memory limit in all three attempts. Fresh Sheet Shrink cleanup followed by the library round trip completed with a median process-creation-to-exit time of 3.021 seconds and a maximum job-commit peak of 891.4 MiB. Prepared Check Performance, Inquire and NXPowerLite outputs completed their library round trips with respective medians of 1.036, 1.054 and 1.101 seconds, and maximum job-commit peaks of approximately 535.6 MiB.

The external prepared-output endpoints omit their tools' preparation cost. The Sheet Shrink endpoint includes fresh cleanup, so these observations do not establish an end-to-end speed ranking among cleaners. Memory-limited attempts remain censored. Recorded commit peaks slightly exceed the configured 4 GiB boundary and are retained without clamping. Concurrent native comparisons and retained filesystem caches qualify this selected-case result. After the timed series, a separate fresh Sheet Shrink control matched all eighteen uncompressed member names, lengths and hashes of the retained native candidate. Its output has 384,724 compressed bytes and 2,581,673 uncompressed member bytes. This representative identity check links the control to the visual audit, while preserving that audit's selected-coverage limitations. The timed trials retain archive hashes, without full per-trial member inventories.

Production cleanup time covers package analysis and output construction. Native preservation-validation cost has its own accounting. A corpus workbook supplies an observed source state. Establishing preservation requires structural checks and independently prepared native controls. Synthetic stress cases vary eligible redundant records, inheritance masks, font witnesses, merged spans and protected references, with fixed-height and automatic-height rows distinguished. Report removed records and bytes together with retained witnesses, exclusions and preservation outcomes under each cleanup configuration.

For each input, report the original and cleaned compressed sizes and their ratio. Report distributions and corpus totals, including inputs with no reduction. Opening time uses a documented readiness signal and fresh owned Excel processes. Single-thread cleanup, cleanup with parallelism and corpus throughput are separate experiments. RAM measurements specify the sampling interval and whether they refer to startup, steady state or peak use.

Microsoft Check Performance is an optional cleanup comparison subject. Its Excel save-only control is necessary to distinguish optimization from serialization and recalculation effects. Its availability does not guarantee that all selected files can be processed. Historical DOM and regular-expression Python implementations have different cleanup decisions. A timing comparison must therefore include output differences and preservation outcomes for each version.

G. Distributed comparator measurements

The native comparison campaigns are distributed across Windows hosts to obtain coverage within the available experimental budget. Native opening, cleanup and saving can dominate the cost of large spreadsheet collections. Each observation therefore retains its host, source identity, configuration and measurement boundary. Raw times remain associated with that host. Cross-tool ratios pair the same original workbook with a Sheet Shrink measurement on the same host. Host-specific ratios retain differences in workload, tool policy and session conditions. Each host and tool remains a separate plotted group. The frozen selection contains the 49 workbooks with the largest absolute compressed XLSX byte savings under the earlier Sheet Shrink screening in each of Destatis, Enron and Data.gov. The timing and preservation outcomes of this extreme-case panel apply to its selected inputs.

The present NXPowerLite lane uses a Lenovo 81YQ laptop with an AMD Ryzen 5 4500U (six physical cores and six

logical processors), 8 GiB installed RAM (7.37 GiB OS-visible), a 512 GB Samsung MZALQ512HALU NVMe SSD and Windows 11 Home, build 26200. The recorded Windows power scheme is Energy Star. Other open Excel and validation processes are retained in the resource receipts. These observations characterize the recorded workstation workload, including any contention.

a) Comparator configuration: NXPowerLite Desktop 11.0.1 x64 uses the custom profile `Cell formatting focused - Top49`. Excess cell-formatting cleanup is enabled. In the Excel profile, image resizing, JPEG conversion, cropped-area deletion, embedded-object flattening, PNG color-depth reduction and embedded chart-data removal are disabled. Lossy image modification conflicts with the objective of exact visible preservation and is deliberately excluded from the comparison policy. The interface retains a JPEG quality control, set to 9. Verification of unchanged images therefore requires the output-level audit.

Consequently, every retained output is audited against the original bytes of media, drawings, charts and embedded objects. Of 147 attempted workbooks, 129 produced reduced packages and ten explicitly remained unchanged. Eight Enron inputs reported an internal NXPowerLite error on both the primary attempt and a diagnostic repeat after restarting the application. Five outputs recompressed PNG images with identical decoded RGBA pixels and image dimensions. A sixth changed an embedded Word document. These six outputs are retained separately and excluded from the formatting-focused timing figure. The remaining 133 outputs passed the protected-part audit. Native worksheet equivalence requires separate visual validation.

XLStylesTool UWP 3.0.1.0 supplies an additional same-host comparison with forced style normalization, name unhiding and wholesale external-name removal disabled. The earlier conversion queue retains 122 hash-matched inputs with a confirmed primary conversion event and its complete per-attempt resource receipt. The other 25 Top49 inputs lack that complete timing evidence and remain outside the timing figure. All 122 included outputs preserve the audited media and object parts byte-for-byte. The joint figure uses the intersection of 110 original workbooks, comprising 21 Destatis, 41 Enron and 48 Data.gov inputs. Both tools therefore use the same original files and repeated Sheet Shrink baselines within each plotted corpus group. The larger individual receipt sets remain available in the artifact. Table V retains the full 133-input NXPowerLite resource lane.

b) Timing and resources: Each included GUI observation has one primary timing measurement per input. Its operational cost limits replication, and its timing variability therefore remains unestimated. Diagnostic retries remain separate attempts. Elapsed time covers automated GUI orchestration through output confirmation and initial package verification. CPU measurements accumulate attributed process-tree user and kernel time. Working-set and private-byte peaks use a nominal 250 ms sampling interval. The existing warm application is reused, with one conversion worker and a 512 MiB RAM and commit reserve. The exact profile and per-attempt evidence accompany the results.

The paired Sheet Shrink experiment uses an optimized release build from revision `765c480`, one active workbook, one Rayon worker and one writer, with the default cleanup policy. One complete warm-up round precedes nine measured rounds over the same 147 source identities. Each round independently shuffles workbook order with a recorded seed. Each file starts a fresh CLI process, and measured wall time ends at process exit. Output hashing is outside this timing boundary. Filesystem caches are left unflushed after warm-up. Direct-child CPU and OS peak-working-set counters are retained, with private usage sampled at a nominal 10 ms polling interval. These memory observations have different sampling semantics from the GUI lane and are reported separately. All 1,470 invocations completed successfully. Each input produced the same package hash across warm-up and measured repetitions, and all 147 final outputs preserved the original bytes of the audited media and object parts. Across inputs, the median relative within-input interquartile range, $(Q_3 - Q_1)/\widetilde{t}$, was 0.84% over the nine measured repetitions.

Figure 10 retains raw times and workbook-level same-host ratios. Each denominator is the median of nine Sheet Shrink trials, and each numerator is the primary comparator observation. Across-workbook quartiles describe heterogeneity in this deliberately selected stress panel. They are distinct from uncertainty over repeated trials [17]. The ratio includes the stated interface and confirmation costs, and the tools have different cleanup policies. It therefore measures the observed cleanup workflows. Opening-time improvements and cleanup-plus-opening costs require their own readiness endpoints, as in the separate historical opening experiment.

H. An external parser failure found during baseline testing

Baseline ingestion exposed an analogous failure in SheetReader. The XLSX conversion of an EDRM Enron workbook terminated its Windows parser while reading `Opt!O306`, a styled empty cell serialized as `<c r="O306" s="105"/>`. Retained value-parser state caused an unsigned length subtraction to underflow before a buffer write. The upstream build terminated in all six targeted trials. With the proposed correction, all six completed over seven worksheets and 24,299 stored values, with stable location/value fingerprints. Trials used one and ten parser threads.

We submitted the correction and synthetic regression fixtures upstream in pull request 5 [18]. The patch also preserves inferred column positions after skipped empty cells. Its fixture suite passed 54 case/thread combinations on Windows x64. The pull request is a submitted contribution, with upstream acceptance pending at this writing. These observations characterize the tested revision and input. They illustrate the need to verify parser transitions and value positions before interpreting ingestion timings.

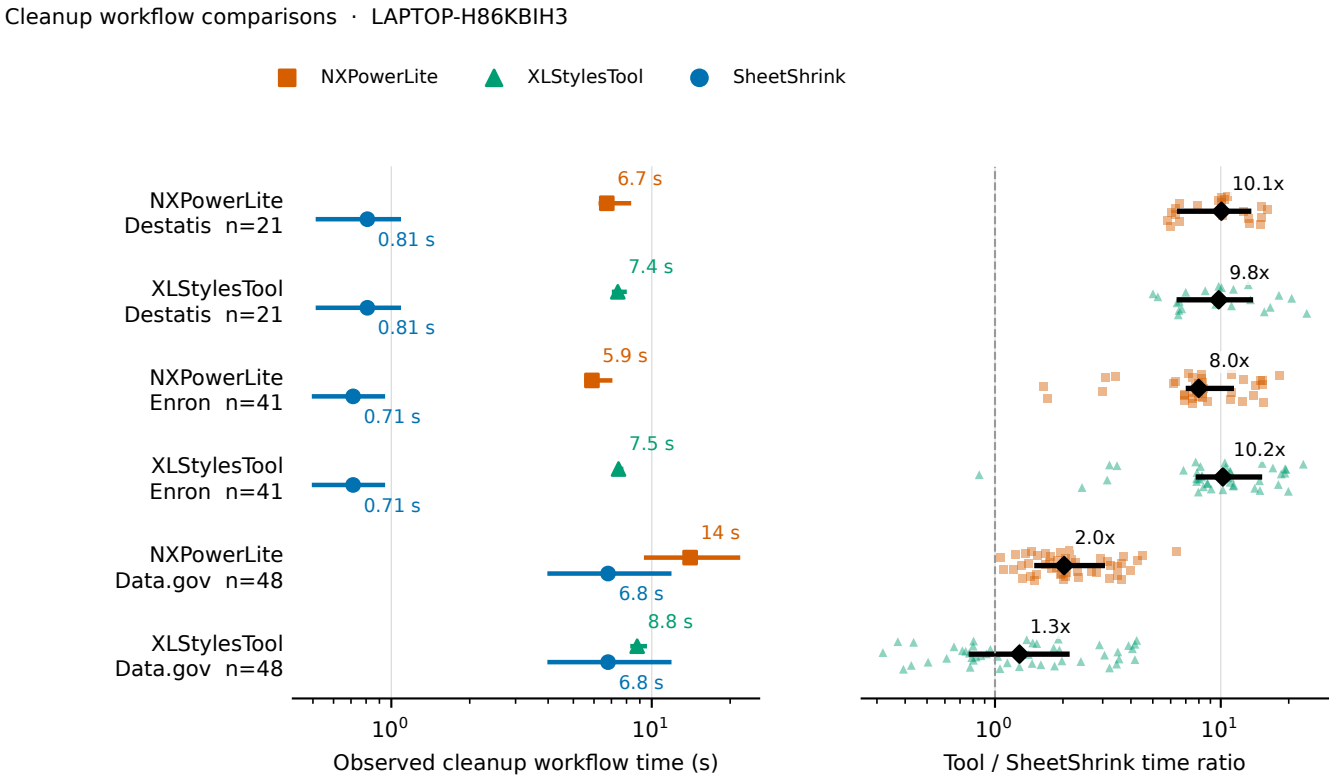


Fig. 10. Same-host NXPowerLite and XLStylesTool cleanup workflow observations on the same 110 originals, paired with Sheet Shrink. Corpus labels give the shared input counts for each tool. Left, absolute times summarized across paired workbooks. Right, workbook-level comparator time divided by the median Sheet Shrink time, on a logarithmic scale. A ratio above one indicates a shorter observed Sheet Shrink workflow. Interval marks show the interquartile range across workbooks. Failures and changed protected payloads remain in the outcome accounting.

I. Six-file cleanup stress panel

A separate stress panel selects six Destatis workbooks from a 100-file historical-reduction screening, including the Migration workbook's hidden zero tail. Each cleaner receives a separate hash-verified original copy. Sheet Shrink uses revision `a7eef1a`. The comparison subjects are NXPowerLite Desktop 11.0.1 with its Balanced profile, native Excel Check Performance in Excel 16.0.20527.20150, and XLStylesTool UWP 3.0.1.0 with optional style normalization, name unhiding and wholesale external-name removal disabled. All 18 external cleanup operations produced retained XLSX outputs with valid ZIP CRCs. Table VI reports compressed package sizes. The artifact records full member names, input/output hashes and exact settings. This historical Balanced lane permits image changes. Its observations remain separate from the subsequent preservation-oriented Top49 profile.

The Migration workbook contains 343,833 numeric zero constants in hidden columns of worksheet `Tabelle 3.4`. NXPowerLite, Check Performance and XLStylesTool retain all of them. Sheet Shrink removes these constants under its unreferenced-hidden-content policy, reducing that worksheet from 361,086 to 787 stored cells while retaining the hidden-column interval. This package observation distinguishes the cleanup footprints even when package reduction alone appears similar.

Native rendering also exposes formatting witnesses. In the Doctors workbook, NXPowerLite removes empty 11-point-font cells Q36:AB36 from worksheet `10.2 T1`. The stored automatic-height row remains, but rebuilt Excel measures row 36 at 12 points instead of the original 15 points. An independent `A1:AC41` repeat retains identical original images across fresh sessions. Original against original has zero differing pixels. NXPowerLite and Check Performance differ in 47,383 and 34,762 pixels, respectively, inside the common image rectangle. Sheet Shrink has zero differing common pixels and a one-pixel white bottom-extent difference. This diagnostic was selected after observing the initial finding.

For Migration, a separate original-derived reference plan covers 48 rectangles over the visible value/formula envelopes of its 36 worksheets. NXPowerLite and Check Performance change overlapping pixels in two and one rectangles, respectively. Sheet Shrink and XLStylesTool preserve common pixels in all 48, with white extent differences accounted for separately. The repeated original image identities agree

across all four subjects. Formatting-only exterior tails, overflow beyond supplied rectangles and outlying object effects remain outside this reference scope. A targeted Traffic attempt exceeded available memory, and targeted Migration attempts encountered full-grid and idle-progress limits. Their failures remain recorded alongside bounded reference follow-ups. These observations describe the selected stress panel. Corpus prevalence, exhaustive rendering coverage and matched execution costs remain separate measurements.

J. Automatic-height regression and correction

The Traffic workbook supplies a concrete counterexample to promotion based only on repeated font formatting [7]. On worksheet `3.8_(1)`, row 105 originally inherits a 12.75-point height without an explicit height or row style. The earlier cleanup promotes its dense empty-cell format into the row. Excel then measures 9 points, moving the following footer. The independently supplied `A1:U110` reference rectangle contains 3,408 changed pixels in the common image area, and its candidate image is five pixels shorter. This is a native presentation failure with retained input, producer and image identities.

Requiring an explicit custom height before promotion restores the original 12.75-point row. Fresh rebuilt original and corrected images both measure 1,358 by 1,327 pixels and are byte-identical. The corrected workbook occupies 2,654,536 bytes against a 37,207,883-byte original. The retained development cleanup took 13.28 seconds with statistics enabled. This single observation does not estimate guard overhead. The source change checks two existing parsed attributes before promotion and retains fixed-height promotion coverage in the regression suite. This correction postdates the six-file panel's pinned `a7eef1a` cleaner and therefore does not retroactively replace its outputs.

A separate current-cleaner Migration repeat produced 1,804,772 bytes from the 16,926,818-byte source. Under DeltaSieve policy v12, 37 image pairs had no pixel differences. A former 12.75-point discrepancy in a zero-width range disappeared when its empty lower context row was excluded. Separate effective-height observations on two other sheets remain in the report. Consequently this repeat supports the recorded image scope and the corrected exclusion, with those remaining observations explicitly retained [19].

K. Preparation and recovered inputs

Eleven initially unsuccessful native preparation cases later completed through visible interaction. Eight used their frozen original inputs. Two required Excel repair and one required resolution of a conflicting defined name. Repair-only baselines identify these three separately. All thirteen retained output and baseline archives passed ZIP integrity checks, while visual validation remains pending. One input first failed on a chart sheet and succeeded after selecting a worksheet. Another completed after a multi-minute suggestion scan. These bounded recovery observations prevent attributing an automation failure to tool incapability. Interactive recovery costs remain separate from automated timings.

L. Validation cost of redundant background definitions

The Destatis workbook `anbau-gartenbaugewaechse-2030224169005.xlsx` contains repeated column-width records over a uniform full-width background. Rebuilt original and Sheet Shrink outputs contain 128,912 and 2,197 column definitions, respectively. A corrected DeltaSieve background-envelope guard reduced the native selection from 16,101 to 165 regions. All 165 image pairs were identical in the fresh comparison. This observation supports the recorded targeted scope. It does not replace the remaining corpus validation or establish exhaustive visual preservation.

a) Explicit no-fill font representatives.: The Destatis workbook `pruefungen-hochschulen-2110420217005.xlsx` exposed a reproducible text-baseline regression on sheet `TAB-04`. An original-versus-original reference control was pixel-identical, whereas the transformed output repeatedly changed 25,640 pixels despite equal capture geometry. Restoring 87 removed blank cells with an explicit default font and no-fill metadata recovered the original image hash. The correction allows this metadata combination to supply the existing per-row font representative, including fixed-height rows with content. A fresh output from the corrected cleaner matched the original over the independent `A1:L916` reference area. The corpus's conflicting legacy print-name aliases were explicitly normalized for opening, with separate hashes and preserved name definitions.

b) Automatic height without an explicit font-application flag.: In `kulturfinanzbericht-gesamtmaterial-1023003229005.xlsx`, removing the blank cell `Impressum!B16` reduced rebuilt row 16 from 15 to 12.75 points. Its XF references Arial MT 12 without an explicit `applyFont`, whereas the text cell uses Arial 10. The stored `ht="15"` has no `customHeight` and does not freeze automatic sizing. The correction admits this non-default font to the existing per-row representative policy and still removes redundant representatives. A separate native control rebuilt both complete workbooks and captured the preselected `Impressum!A1:J43` area. The 832 by 743 images were byte-identical with equal geometry. This local control completed in 13.07 seconds. A subsequent complete targeted comparison under capture policy version 15 completed all 350 selected workbook regions without a pixel, geometry or rendering difference. Historical findings remain retained with their original planning policy.

M. Historical Inquire workflow observations

A separate 46-input subset fills missing historical Inquire observations from the same stress selection. All 46 repeated Sheet Shrink process measurements completed. The retained Inquire outcomes comprise 24 complete, three partial, 18

failed and one unsupported attempt. Its complete workflows include startup, opening, cleanup, saving and termination. They are compared descriptively with repeated Sheet Shrink process times and have different endpoints and repetition counts. Their median ratio is 13.38 over the 24 complete pairs. Partial and failed attempts receive no ratio. These measurements cannot isolate algorithm execution cost.

N. Targeted visual outcomes on the Top49 selection

A version-matched audit of the 49 workbooks with the largest absolute Sheet Shrink reductions in each corpus retained 42 Destatis, 46 Data.gov, and 46 Enron comparisons against the selected originals. Separately documented normalized baselines supplied five further Destatis and two further Enron observations. The remaining two, three, and one slots, respectively, had no eligible observation. Across both baseline categories, zero Destatis, seven Data.gov, and five Enron workbooks had recorded differences. These counts describe the selected capture scope and include findings whose cause remains unresolved. The five Enron workbooks had six chart findings of 2–26 changed pixels with equal image dimensions and no reported geometry change. The missing Enron workbook required repair when opened in Excel, including after a separate legacy reconversion.

O. Evidence boundary

The selected controls demonstrate concrete preservation conditions and recorded failures. They do not estimate corpus-wide defect prevalence. Final corpus reduction, matched cleanup-plus-opening cost and independent-reference coverage remain open. Detailed development attempts, input hashes, native images and producer versions remain in the source-linked artifact. The main comparison separates those receipts from the reported completed observations.

TABLE II

OPENING/MEMORY CASE-STUDY INPUTS AND NATIVE VISUAL EVIDENCE. SIZES ARE DECIMAL MB. PAYLOAD IS THE SUM OF ALL UNCOMPRESSED ZIP-MEMBER SIZES. REDUCTION COLUMNS COMPARE EACH OUTPUT WITH ITS ORIGINAL. PENDING IDENTIFIES UNAVAILABLE EVIDENCE. PAIRED BARS SHOW ORIGINAL (GRAY) AND OUTPUT (BLUE/ORANGE), WITH ORIGINAL = 100%.

Variant	ZIP (reduction)	Payload (reduction)	Pixels D/N	Sheets	Geom.	State
Destatis Migration						
Original	16.927 (0%)	248.39 (0%)	Ref.	Ref.	Ref.	Ref.
Sheet Shrink	1.805 (89%)	Pending	Pending	Pending	Pending	Pending
NXPowerLite	11.741 (31%)	197.48 (20%)	Pending	Pending	Pending	M
XLStylesTool	16.923 (0.022%)	248.42 (-0.0095%)	Pending	Pending	Pending	M
Check Performance	1.817 (89%)	20.15 (92%)	Pending	Pending	Pending	M
Inquire	9.499 (44%)	116.44 (53%)	Pending	Pending	Pending	M
Destatis Municipal Finance						
Original	89.367 (0%)	955.41 (0%)	Ref.	Ref.	Ref.	Ref.
Sheet Shrink	4.603 (95%)	Pending	Pending	Pending	Pending	Pending
NXPowerLite	2.603 (97%)	21.60 (98%)	34,496 / Pending	37/82/107	1	L
XLStylesTool	89.362 (0.0056%)	955.40 (0.0017%)	Pending	Pending	Pending	M
Check Performance	2.817 (97%)	21.20 (98%)	405,929 / Pending	43/78/107	1	L
Inquire	34.155 (62%)	738.21 (23%)	Pending	Pending	Pending	M
Data.gov QT/RR						
Original	121.949 (0%)	1350.77 (0%)	Ref.	Ref.	Ref.	Ref.
Sheet Shrink	3.848 (97%)	Pending	Pending	Pending	Pending	Pending
NXPowerLite	3.331 (97%)	29.98 (98%)	Pending	Pending	Pending	M
XLStylesTool	121.949 (1.9e-05%)	1350.77 (-7.8e-05%)	Pending	Pending	Pending	M
Check Performance	3.626 (97%)	25.59 (98%)	Pending	Pending	Pending	F
Inquire	3.878 (97%)	30.13 (98%)	Pending	Pending	Pending	M
Enron b2d0dfef						
Original	42.711 (0%)	353.15 (0%)	Ref.	Ref.	Ref.	Ref.
Sheet Shrink	Pending	Pending	Pending	Pending	Pending	Pending
NXPowerLite	Pending	Pending	Pending	Pending	Pending	Pending
XLStylesTool	Pending	Pending	Pending	Pending	Pending	Pending
Check Performance	Pending	Pending	Pending	Pending	Pending	Pending
Inquire	Pending	Pending	Pending	Pending	Pending	Pending

Sheet Shrink outputs are freshly generated in each six-worker trial. Their compressed size is recorded consistently across repetitions, while their uncompressed payload and matched native visual evidence remain pending. NXPowerLite uses the cell-formatting-focused profile. XLStylesTool retains the tested settings. Ref. identifies the original baseline. L denotes completed targeted captures with coverage limitations, M a memory-limit stop, F a failed attempt and Pending unavailable evidence. Sheets gives changed/observed/total worksheets. Geometry counts flagged capture findings. Pixel counts aggregate retained diagnostic observations under policy v12 after paired native rebuild. The total comparable-pixel denominator N remains pending. Different capture scopes and unresolved attribution constrain comparisons. Gray bars denote the original. Blue denotes a value at or below the original, and orange a larger value. Orange cell backgrounds mark observed visual differences. Each pair has a common zero. Original = 100%, with a shared linear scale within each workbook and size column. Absolute MB and reduction percentages accompany the bars.

TABLE III

OPENING/SAVING RESOURCES ON THE RYZEN 5 4500U LAPTOP, 8 GIB INSTALLED RAM, SIX LOGICAL PROCESSORS. DESTATIS MIGRATION. PART 1 OF 3. EACH ROW HAS THREE ATTEMPTS UNDER A 4 GIB JOB-COMMIT CAP AND 900 S TIMEOUT.

Variant	Cleanup	Open	Save	End-to-end	RSS lib.	RSS all	Commit	Outcome
		seconds, median of completed trials			MiB, maximum across attempts			
openpyxl								
Original	–	53.74	53.59	111.30	2700	2700	2881	3/3
Sheet Shrink	1.46	3.89	3.29	9.67	250	377	576	3/3
NXPowerLite	prep.	39.99	40.75	83.58	1996	1996	2175	3/3
XLStylesTool	prep.	53.54	53.13	111.13	2700	2700	2881	3/3
Check Performance	prep.	4.25	3.70	9.11	236	236	411	3/3
Inquire	prep.	26.69	25.51	54.96	1530	1530	1721	3/3
pandas to XLSX								
Original	–	Pending	Pending	M@20.84	3900	3900	4096	0/3 M
Sheet Shrink	1.42	4.54	27.92	35.83	856	856	1014	3/3
NXPowerLite	prep.	Pending	Pending	M@20.75	3900	3900	4096	0/3 M
XLStylesTool	prep.	Pending	Pending	M@20.83	3900	3900	4096	0/3 M
Check Performance	prep.	Pending	Pending	M@8.67	3900	3900	4096	0/3 M
Inquire	prep.	Pending	Pending	M@9.33	3900	3900	4096	0/3 M
pandas to CSV								
Original	–	Pending	Pending	M@20.48	3900	3900	4096	0/3 M
Sheet Shrink	1.42	4.50	4.73	11.54	214	419	577	3/3
NXPowerLite	prep.	Pending	Pending	M@20.72	3900	3900	4096	0/3 M
XLStylesTool	prep.	Pending	Pending	M@20.55	3900	3900	4096	0/3 M
Check Performance	prep.	Pending	Pending	M@8.48	3900	3900	4096	0/3 M
Inquire	prep.	Pending	Pending	M@9.30	3900	3900	4096	0/3 M
Excel COM								
Original	–	8.69	9.10	20.34	429	429	304	3/3
Sheet Shrink	1.41	1.94	4.15	9.92	309	419	606	3/3
NXPowerLite	prep.	7.16	8.29	17.94	409	409	282	3/3
XLStylesTool	prep.	8.62	9.02	20.17	428	428	304	3/3
Check Performance	prep.	2.08	0.64	5.34	259	259	140	3/3
Inquire	prep.	4.66	2.75	9.99	324	324	195	3/3
R openxlsx								
Original	–	15.08	28.78	44.38	824	824	835	3/3
Sheet Shrink	1.41	4.41	7.35	13.70	317	421	606	3/3
NXPowerLite	prep.	10.39	19.14	30.04	684	684	697	3/3
XLStylesTool	prep.	15.19	29.20	44.92	831	831	848	3/3
Check Performance	prep.	3.95	7.23	11.70	286	286	298	3/3
Inquire	prep.	10.50	21.21	32.22	564	564	578	3/3
R readxl + writexl								
Original	–	Pending	Pending	M@8.48	4070	4070	4096	0/3 M
Sheet Shrink	1.41	175.34	0.87	178.17	262	392	622	3/3
NXPowerLite	prep.	Pending	Pending	M@8.36	4070	4070	4096	0/3 M
XLStylesTool	prep.	Pending	Pending	M@8.40	4070	4070	4096	0/3 M
Check Performance	prep.	Pending	Pending	M@7.59	4072	4072	4096	0/3 M
Inquire	prep.	Pending	Pending	M@7.67	4072	4072	4096	0/3 M

Sheet Shrink runs fresh cleanup immediately before the library. Its end-to-end clock spans cleanup launch through library exit. Original and prepared NXPowerLite, XLStylesTool, Check Performance and Inquire rows report library process lifetime. prep. denotes preparation outside this trial. Open and save measure the retained operation endpoints. End-to-end includes startup/shutdown and is measured directly. RSS lib. is the maximum sampled process-tree working set in importing, opening, saving and completed phases. RSS all includes cleanup and other recorded phases. Sampling is nominally 20 ms. Commit is the Windows job peak. The columns have different counter semantics. Outcome gives completed/attempted. M@t is elapsed time until the memory-limit stop, with missing completion phases marked Pending. Blue/orange bars indicate values at or below/above the original on the same host and endpoint. Paired bars in end-to-end and library RSS cells use original = 100%, a common zero and a shared linear scale per library/workbook/host. Gray denotes the original and blue/orange the compared value. Bars require three completed attempts in both conditions. Other hosts, dispersion and CPU fields remain explicit in the machine-readable artifact.

TABLE III

OPENING/SAVING RESOURCES ON THE RYZEN 5 4500U LAPTOP, 8 GIB INSTALLED RAM, SIX LOGICAL PROCESSORS. DESTATIS MUNICIPAL FINANCE. PART 2 OF 3. EACH ROW HAS THREE ATTEMPTS UNDER A 4 GIB JOB-COMMIT CAP AND 900 S TIMEOUT.

Variant	Cleanup	Open	Save	End-to-end	RSS lib.	RSS all	Commit	Outcome
		seconds, median of completed trials			MiB, maximum across attempts			
openpyxl								
Original	–	Pending	Pending	M@79.56	3914	3914	4097	0/3 M
Sheet Shrink	5.95	7.98	9.05	24.39	452	1061	1446	3/3
NXPowerLite	prep.	4.45	6.10	11.63	227	227	404	3/3
XLStylesTool	prep.	Pending	Pending	M@79.95	3914	3914	4097	0/3 M
Check Performance	prep.	4.28	6.04	11.40	227	227	404	3/3
Inquire	prep.	Pending	Pending	M@112.79	3913	3913	4097	0/3 M
pandas to XLSX								
Original	–	183.67	8.02	192.94	668	668	836	3/3
Sheet Shrink	5.38	6.47	7.75	20.70	231	1082	1452	3/3
NXPowerLite	prep.	3.39	7.65	12.21	210	210	375	3/3
XLStylesTool	prep.	184.52	7.73	193.39	668	668	835	3/3
Check Performance	prep.	3.41	7.77	12.39	210	210	375	3/3
Inquire	prep.	68.09	7.61	76.89	621	621	788	3/3
Excel COM								
Original	–	27.74	11.98	42.46	1000	1000	889	3/3
Sheet Shrink	5.36	2.40	1.32	69.76	412	1183	1450	3/3
NXPowerLite	prep.	2.04	0.66	63.05	388	388	284	3/3
XLStylesTool	prep.	27.91	11.98	100.28	1000	1000	889	3/3
Check Performance	prep.	1.59	0.55	62.47	252	252	139	3/3
Inquire	prep.	23.27	10.70	35.87	616	616	502	3/3
R openxlsx								
Original	–	46.10	55.99	102.66	2115	2115	2164	3/3
Sheet Shrink	5.44	3.72	8.87	18.56	269	1117	1446	3/3
NXPowerLite	prep.	3.00	8.20	11.67	273	273	286	3/3
XLStylesTool	prep.	46.03	55.50	102.17	2012	2012	2030	3/3
Check Performance	prep.	2.63	7.35	10.52	268	268	285	3/3
Inquire	prep.	19.17	33.29	53.09	1246	1246	1257	3/3
R readxl + writexl								
Original	–	21.35	0.90	22.71	1906	1906	2115	3/3
Sheet Shrink	5.43	14.83	0.86	21.62	242	1059	1446	3/3
NXPowerLite	prep.	14.97	0.85	16.26	223	223	249	3/3
XLStylesTool	prep.	21.36	0.88	22.65	1906	1906	2115	3/3
Check Performance	prep.	14.78	0.86	16.08	221	221	247	3/3
Inquire	prep.	18.98	0.89	20.33	668	668	757	3/3

Sheet Shrink runs fresh cleanup immediately before the library. Its end-to-end clock spans cleanup launch through library exit. Original and prepared NXPowerLite, XLStylesTool, Check Performance and Inquire rows report library process lifetime. prep. denotes preparation outside this trial. Open and save measure the retained operation endpoints. End-to-end includes startup/shutdown and is measured directly. RSS lib. is the maximum sampled process-tree working set in importing, opening, saving and completed phases. RSS all includes cleanup and other recorded phases. Sampling is nominally 20 ms. Commit is the Windows job peak. The columns have different counter semantics. Outcome gives completed/attempted. M@t is elapsed time until the memory-limit stop, with missing completion phases marked Pending. Blue/orange bars indicate values at or below/above the original on the same host and endpoint. Paired bars in end-to-end and library RSS cells use original = 100%, a common zero and a shared linear scale per library/workbook/host. Gray denotes the original and blue/orange the compared value. Bars require three completed attempts in both conditions. Other hosts, dispersion and CPU fields remain explicit in the machine-readable artifact.

TABLE III

OPENING/SAVING RESOURCES ON THE RYZEN 5 4500U LAPTOP, 8 GIB INSTALLED RAM, SIX LOGICAL PROCESSORS. DATA.GOV QT/RR. PART 3 OF 3. EACH ROW HAS THREE ATTEMPTS UNDER A 4 GIB JOB-COMMIT CAP AND 900 S TIMEOUT.

Variant	Cleanup	Open	Save	End-to-end	RSS lib.	RSS all	Commit	Outcome
		seconds, median of completed trials			MiB, maximum across attempts			
openpyxl								
Original	–	Pending	Pending	M@79.42	3913	3913	4136	0/3 M
Sheet Shrink	7.65	5.83	5.19	19.84	318	1600	2079	3/3
NXPowerLite	prep.	5.93	5.18	12.26	299	299	475	3/3
XLStylesTool	prep.	Pending	Pending	M@78.69	3913	3913	4136	0/3 M
Check Performance	prep.	5.55	5.26	12.07	299	299	475	3/3
Inquire	prep.	5.91	5.22	12.39	303	303	478	3/3
pandas to XLSX								
Original	–	237.47	10.20	249.13	327	327	493	3/3
Sheet Shrink	7.03	4.53	10.14	22.99	346	1670	2080	3/3
NXPowerLite	prep.	4.51	10.13	15.79	327	327	493	3/3
XLStylesTool	prep.	235.07	10.24	246.68	327	327	493	3/3
Check Performance	prep.	4.27	10.15	15.55	327	327	493	3/3
Inquire	prep.	4.54	10.31	16.09	327	327	493	3/3
Excel COM								
Original	–	32.42	4.59	39.79	929	929	816	3/3
Sheet Shrink	7.10	1.74	0.57	11.84	272	1741	2092	3/3
NXPowerLite	prep.	1.68	0.56	4.67	253	253	136	3/3
XLStylesTool	prep.	31.99	4.50	39.02	931	931	816	3/3
Check Performance	prep.	1.62	0.53	62.52	246	246	133	3/3
Inquire	prep.	1.71	0.53	4.77	247	247	133	3/3
R openxlsx								
Original	–	63.19	75.45	140.86	2453	2453	2593	3/3
Sheet Shrink	7.03	1.48	2.32	11.28	242	1744	2106	3/3
NXPowerLite	prep.	1.44	2.32	4.26	224	224	236	3/3
XLStylesTool	prep.	63.10	76.70	140.40	2454	2454	2593	3/3
Check Performance	prep.	1.15	1.87	3.55	210	210	223	3/3
Inquire	prep.	1.47	2.33	4.32	225	225	237	3/3
R readxl + writexl								
Original	–	10.48	1.02	11.90	1341	1341	1492	3/3
Sheet Shrink	7.05	1.57	1.01	10.06	258	1730	2163	3/3
NXPowerLite	prep.	1.53	1.00	2.94	240	240	256	3/3
XLStylesTool	prep.	10.45	1.02	11.89	1340	1340	1492	3/3
Check Performance	prep.	1.49	0.97	2.88	248	248	271	3/3
Inquire	prep.	1.55	1.01	2.96	239	239	256	3/3
Enron fourth-case measurements use a separate host lane in Section IV-F.								

Sheet Shrink runs fresh cleanup immediately before the library. Its end-to-end clock spans cleanup launch through library exit. Original and prepared NXPowerLite, XLStylesTool, Check Performance and Inquire rows report library process lifetime. prep. denotes preparation outside this trial. Open and save measure the retained operation endpoints. End-to-end includes startup/shutdown and is measured directly. RSS lib. is the maximum sampled process-tree working set in importing, opening, saving and completed phases. RSS all includes cleanup and other recorded phases. Sampling is nominally 20 ms. Commit is the Windows job peak. The columns have different counter semantics. Outcome gives completed/attempted. M@t is elapsed time until the memory-limit stop, with missing completion phases marked Pending. Blue/orange bars indicate values at or below/above the original on the same host and endpoint. Paired bars in end-to-end and library RSS cells use original = 100%, a common zero and a shared linear scale per library/workbook/host. Gray denotes the original and blue/orange the compared value. Bars require three completed attempts in both conditions. Other hosts, dispersion and CPU fields remain explicit in the machine-readable artifact.

TABLE IV

SELECTED-CASE FOLLOW-UP UNDER A 24 GIB JOB-COMMIT LIMIT. ELAPSED TIME COVERS PROCESS CREATION THROUGH EXIT, INCLUDING FRESH CLEANUP IN THE PIPELINE. TIMES ARE MEDIANS OF THREE ATTEMPTS. COMMIT PEAKS ARE MAXIMA OVER THE THREE ATTEMPTS. M DENOTES THREE MEMORY-LIMIT STOPS. THE RAW RECEIPTS RETAIN RANGES, PHASE SAMPLES, WORKING SETS AND CPU COUNTERS.

Workbook / endpoint	Original (s)	Cleanup + endpoint (s)	Ratio	Original peak (GiB)	Pipeline peak (GiB)
Migration / pandas to XLSX	M	24.729	–	24.00	1.27
Migration / pandas to CSV	M	8.291	–	24.00	0.65
Migration / readxl + writexl	392.149	131.953	2.97	8.70	0.51
Finance / openpyxl	255.539	15.526	16.46	8.07	2.62
QT/RR / openpyxl	279.638	11.519	24.28	12.56	3.66

TABLE V

MEDIAN RESOURCE OBSERVATIONS ACROSS PAIRED WORKBOOKS ON THE LAPTOP. CPU IS PROCESS TIME IN SECONDS. WS IS WORKING SET IN MIB, USING OS PEAK COUNTERS FOR SHEET SHRINK AND SAMPLED PEAKS FOR NXPOWERLITE.

Corpus	Pairs	CPU Sheet Shrink	CPU NXPowerLite	WS Sheet Shrink	WS NXPowerLite
Destatis	44	1.24	5.45	54	248
Enron	41	0.70	4.19	51	306
Data.gov	48	6.73	10.49	415	146

TABLE VI

SELECTED DESTATIS STRESS PANEL. ORIGINAL SIZE IS DECIMAL MB. OTHER COLUMNS GIVE PACKAGE REDUCTION (%).

Workbook	Original	Sheet Shrink	NXPowerLite	Check Performance	XLStylesTool
Finance	89.37	94.85	97.09	96.85	0.01
Traffic	37.21	92.99	65.83	94.04	0.02
Migration	16.93	90.67	30.64	89.27	0.02
Electricity	1.43	89.86	91.92	63.96	0.11
Doctors	15.24	78.22	79.52	82.00	0.02
Time use	3.84	81.38	83.14	81.66	0.32

V. Related Work

A. Workbook bloat and cleanup tools

Microsoft Check Performance removes cells associated with unnecessary formatting and other stored information from a loaded workbook [20]. Its documentation warns that optimization can remove pixel-art cells. Inquire's Clean Excess Cell Formatting command removes excess cells beyond the last nonblank cell and can change conditional-formatting coverage [21]. O'Beirne's published analysis of Excel 2013 Inquire discusses this command and criticizes its lack of information about planned and completed removals [22]. This historical analysis establishes published attention to cleanup and its transparency. Current product behavior requires version-specific evidence. These documented policies motivate comparing both reduction and preservation outcomes under explicit contracts.

Package-level cleanup also predates Sheet Shrink. XLStylesTool's developer describes an Open-XML-based cleaner for excessive and corrupted styles that operates independently of Excel's object model [23]. NXPowerLite Desktop 10 lists excess XLSX cell-format cleanup among its features [24]. Its engine description discusses formatted blank cells as a source of large worksheet parts [25]. These are relevant comparison subjects. Their available descriptions do not establish the guarded removal rules, compact row-font witnesses or native preservation outcomes examined here. Those properties require version-specific evidence. Sheet Shrink's positioning therefore includes its effect-aware rules, retained representatives and measured costs alongside its execution before native loading. Its removal decisions use persisted OOXML properties and workbook context. The cleanup phase has no native rendering observations or measured automatic row heights available to guide deletion or reconstruct layout. Small retained font and inheritance witnesses must preserve these effects when Excel later opens the result. Native validation supplies subsequent experimental evidence for the decisions described in the method.

B. Spreadsheet scalability and system optimization

Mack et al.'s CHI Extended Abstracts case study characterizes scalability problems reported in spreadsheet forums [26]. Rahman et al. benchmark Excel, LibreOffice Calc and Google Sheets on spreadsheet computations and examine indexing, data layout, incremental evaluation and shared computation [27]. These studies motivate performance work on spreadsheets. Their workloads assess system responsiveness and computation, while Sheet Shrink measures removable persisted representation and the cost of preparing workbooks for subsequent use.

DataSpread's storage engine integrates spreadsheet presentation with database storage and positional access [28]. Anti-Freeze schedules asynchronous formula computation to retain interactivity during updates [29]. Hillview distributes exploration of large datasets and uses compact visualization sketches [30]. TACO exploits regular formula-reference patterns to compact dependency graphs and query them directly [31]. These systems optimize storage, computation or exploration within a processing system. Sheet Shrink transforms an existing XLSX package into another editable workbook under a stated preservation contract. These optimization layers can be used together.

C. Spreadsheet loading and specialized parsing

SheetReader studies specialized spreadsheet ingestion and the coupling of parsing, decompression and parallel execution [1]. It motivates evaluating loading cost and memory. Its output consists of data representations for an analytical consumer. Sheet Shrink produces an editable workbook whose required layout and dependencies must survive package rewriting. A blank cell that a tabular reader can omit may still influence inherited formatting or automatic layout in Excel. Our evaluation therefore measures cleanup together with subsequent use of the output and separately observes native preservation. The two systems address complementary sources of ingestion cost, namely the reader's processing strategy and the representation supplied to it.

D. Spreadsheet refactoring and validation

Hermans et al. detect formula smells and suggest refactorings to improve spreadsheet maintainability [32]. Their work connects structural changes to behavioral obligations. Sheet Shrink's removal rules additionally account for formatting inheritance and native layout effects of stored cells.

Spreadsheet slicing integrates dependency reasoning with testing and debugging [33]. Dependencies also constrain removal when retained formulas or graphical objects refer to candidate content. Workbook comparison tools provide observations about cell and format differences [34]. Our native validation protocol uses DeltaSieve's region selection and capture workflow [10], with independent references and mutation controls to evaluate selected coverage. These validation mechanisms provide evidence about cleanup outcomes after the package transformation.

VI. Discussion and Limitations

The preservation contract is narrower than all possible spreadsheet behavior. Rendering depends on the installed Excel version, fonts and calculation state. A selected-region observation has limited scope. Unsupported XML and unsuccessful opening or saving remain visible outcomes. The independent reference sample estimates behavior on its eligible population and cannot establish a universal absence of regressions.

Large reductions can coexist with small preservation witnesses. Font metadata, inheritance masks and content-bearing hidden tails therefore require separate conditions. The constant-tail policy remains a structural heuristic. The fast CRC/length identity shortcut is collision-prone, and the selected verification mode must accompany measurements. The built-in-name predicate also lacks a formula-body equivalence check. These limitations constrain preservation claims.

Regression cases can support a reusable methodological observation when their mechanism and control are retained. The Traffic example identifies a failure of automatic-height style promotion, while the Migration example identifies an exterior-context classification error in the validator. Their remedies operate at different stages. Preserving that distinction prevents a validator change from being reported as a cleaner improvement. Operational failures in the native comparison environment also affect achieved evaluation coverage and belong in the recorded outcomes even when cleanup itself completed quickly.

VII. Conclusion

Sheet Shrink addresses spreadsheet redundancy through effect-aware removal and compact preservation. Its implementation and evaluation distinguish compressed size, XML changes, execution cost and observed presentation. The completed manuscript must quantify those outcomes using the frozen experiments described above and state the remaining unsupported cases.

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OpenAI ChatGPT and Codex assisted with drafting and editing the Abstract and Sections I–VII, literature discovery and organization, implementation and refactoring, test and experiment scripts, figure-generation code, and analysis of retained experiment evidence. This assistance included the method diagrams, XML examples and execution visualizations. Local agent workflows using these systems also supported overnight validation campaigns by inspecting status and logs, investigating stalled attempts, and documenting candidate visual regressions for author review.

The author directed development, reviewed source code, and performed manual testing and debugging. Agent observations supported investigation. Reported validation outcomes are grounded in the retained native captures and comparison records. The author remains responsible for the scientific claims, references, evidence selection, originality and final submitted content.

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